

LJM: A Latent Joint-conditional Model for Action-Conditioned World Model

Anonymous submission

Abstract

Action-conditioned embodied world models aim to predict how robot scenes evolve under future control commands, yet existing video-generation approaches often inject low-dimensional action signals directly into diffusion backbones. Such a simple conditioning scheme is often insufficient for manipulation, where future visual outcomes depend heavily on the current scene and interaction state. To better model this dependence between actions and scene dynamics, we present **LJM**, a Latent Joint-conditional Model that transforms language, visual context, and future actions into action-aware latent tokens before world modeling. A latent reasoning expert, supervised by future visual, proprioceptive, and optical-flow targets, is coupled with a video diffusion expert through Mixture-of-Transformers (MoT) Shared Attention, establishing an explicit interface between robot control and future visual dynamics. We evaluate LJM in terms of both generation quality and downstream policy utility. On WorldArena, LJM achieves 73.06 EWMScore, outperforming the strongest baseline by 3.47 points. Across RoboTwin bimanual and real-world Franka tasks, LJM accurately ranks representative policies and achieves a Pearson correlation of 0.785 with real performance, enabling reliable closed-loop policy evaluation. Moreover, by converting failure trajectories into successful synthetic trajectories for supervised fine-tuning, LJM improves real-world policy success from 35% to 90%.

Introduction

Action-conditioned world models (ACWMs) aim to predict how visual scenes evolve under sequences of robot actions. By providing an action-controllable proxy for physical interaction, a capable ACWM can reduce dependence on costly robot rollouts and support robotic video prediction, policy evaluation, and policy improvement (Wang et al. 2026b; Quevedo et al. 2025; Li et al. 2025; Jiang et al. 2025). However, the utility of an ACWM depends on more than generating photorealistic videos. It must faithfully capture how commanded robot motions alter the joint state of the robot and its environment, such that the predicted visual future remains consistent with both the action sequence and the ongoing interaction. This requirement is particularly important for manipulation: inaccurate robot motion or object responses not only degrade video quality, but can also produce misleading policy evaluations and unreliable learning signals for policy improvement.

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Most existing ACWMs condition video diffusion models by directly injecting low-dimensional robot actions through linear projections, cross-attention, or FiLM/AdaLN-style modulation (Guo et al. 2025; Wang et al. 2026a). Such conditioning informs the video generator of the commanded robot motion, but does not explicitly represent the resulting robot–environment interaction. For example, the same delta end-effector motion should produce fundamentally different visual outcomes before contact, during pushing, and after grasping an object. Although these interaction states may be inferred from the current observation, existing approaches leave the video diffusion backbone to discover their action-dependent consequences implicitly from the video-generation objective. Consequently, a model may generate visually plausible futures while producing incorrect robot or object dynamics—for instance, moving an object before contact, failing to move it after grasping, or generating an end-effector trajectory inconsistent with the commanded action. This action–consequence gap suggests that reliable world modeling requires more than directly conditioning video generation on actions: the model should explicitly predict how the robot and environment will jointly evolve before rendering the future video.

In this paper, we present **LJM**, a Latent Joint-conditional Model that introduces a reason-before-render paradigm for action-conditioned world modeling. Inspired by latent reasoning for robotic action generation (Liu et al. 2026), LJM uses a latent reasoning expert to predict how the robot and environment will evolve under a given future action sequence before synthesizing the corresponding video. Specifically, conditioned on the language instruction, current visual observation, and future actions, the reasoning expert predicts temporally structured robot–environment interaction latent tokens in a single causal-masked forward pass for efficient inference. These tokens are supervised by three complementary types of future information: visual latents describe how the scene changes, optical-flow latents capture how the robot and objects move, and proprioceptive latents represent how the robot state evolves. These future targets are used only during training; at inference time, LJM predicts the interaction tokens solely from the available observation, instruction, and actions. The resulting latents are then shared with a video diffusion expert to guide the generation of action-consistent future frames.

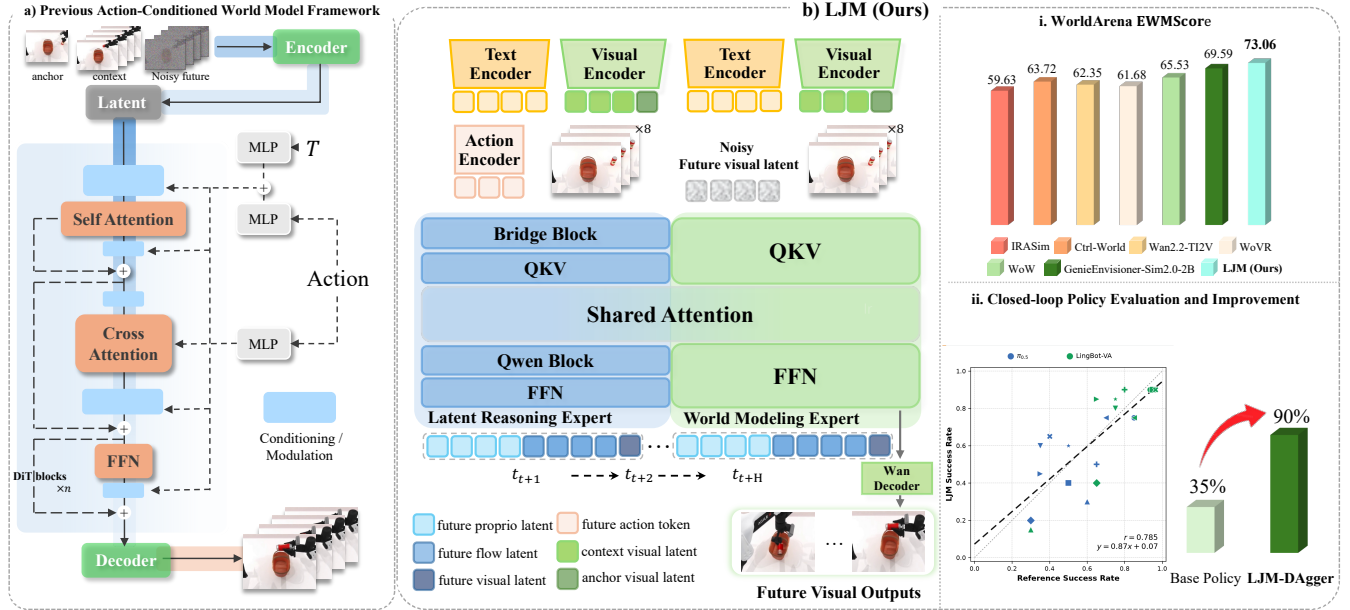


Figure 1: **Comparison between previous action-conditioned world models and LJM.** The left side illustrates the architectural difference: prior methods condition video diffusion models by directly injecting robot actions through lightweight adapters, whereas LJM first transforms language, visual context, and future actions into action-aware interaction latent tokens using a latent reasoning expert, and then couples these latents with the video diffusion expert via MoT Shared Attention. The right side presents that LJM achieves the highest WorldArena EWMScore among compared world models, while enabling substantial improvement in downstream closed-loop policy performance through LJM-Dagger.

To integrate latent interaction reasoning with visual synthesis, LJM adopts a Mixture-of-Transformers architecture that couples a vision-language Transformer with a video diffusion Transformer. The latent reasoning expert converts the instruction, visual context, and future actions into robot-environment latent tokens, while the world modeling expert denoises future video latents. The two experts maintain specialized parameters but exchange information through shared attention, allowing the predicted interaction dynamics to directly guide future-frame generation. Meanwhile, LJM retains the conventional direct action-conditioning pathway through cross-attention and AdaLN modulation, ensuring that the video expert has direct access to the commanded low-level motion. By jointly optimizing the video-generation objective and the latent prediction objective, LJM explicitly connects robot control, interaction dynamics, and future visual outcomes within a unified model.

We evaluate LJM from both video-generation quality and downstream policy utility. On WorldArena (Shang et al. 2026), LJM achieves an EWMScore of 73.06, ranking second on the complete leaderboard and first among the representative baselines compared in this work. On LIBERO (Liu et al. 2023a), it obtains the best performance among the evaluated methods across all four video-quality metrics, including PSNR, SSIM, FVD, and LPIPS. We further evaluate LJM as a closed-loop proxy environment for $\pi_{0.5}$ (Physical Intelligence, Black et al. 2025) and LingBot-VA (Li et al. 2026) policies across five RoboTwin (Chen et al. 2025) and five real-world Franka tasks. Policy performance measured

through LJM rollouts agrees with reference performance, achieving a Pearson correlation of 0.785 and an average task-wise MMRV of 0.008, while correctly identifying the better-performing policy on nine of the ten tasks. Finally, we use LJM as a data engine by generating failure-conditioned synthetic trajectories. On a challenging Franka task, post-training $\pi_{0.5}$ entirely on LJM-generated data improves real-world success from 35% to 90%, outperforming an equally sized expert-only dataset by 15 percentage points. Our main contributions are summarized as follows:

- We propose **LJM**, a *reason-before-render* action-conditioned world model that couples vision-language reasoning and video diffusion experts through shared attention, augmenting action conditioning with a structured robot-environment interaction latent.
- We design temporally structured **interaction latent tokens**, predicted in a single forward pass and supervised by future visual, optical-flow, and proprioceptive targets to capture scene changes, motion dynamics, and robot-state evolution.
- We evaluate LJM across video generation, policy evaluation, and policy improvement. LJM achieves an EWMScore of 73.06, ranking second on the complete WorldArena leaderboard, and leads all four LIBERO metrics among the evaluated methods. Its rollouts achieve 100% task-wise policy-selection accuracy on RoboTwin, while action-perturbed synthetic rollouts improve policy learning by 55pp over training on the original data alone.

Related Work

Action-Conditioned Embodied World Models

Embodied world models simulate how environments evolve under agent actions, providing predictive interfaces for robotic video generation, policy evaluation, and policy improvement. Recent works have studied action-conditioned world modeling across robotic manipulation and embodied environments. IRASim models fine-grained manipulation dynamics, while EnerVerse-AC and Ctrl-World develop controllable generative world models conditioned on robot actions and scene context (Zhu et al. 2025a; Jiang et al. 2025; Guo et al. 2025). Beyond visual prediction, WorldGym and WorldEval use world models as proxy environments for robot policy evaluation, while Gemini Robotics in a Veo simulator, Interactive World Simulator, WMPO, and WoVR explore their utility for policy training, evaluation, and post-training (Quevedo et al. 2025; Li et al. 2025; Gemini Robotics Team et al. 2025; Wang et al. 2026a; Zhu et al. 2025b; Jiang et al. 2026). WorldArena further evaluates embodied world models as policy evaluators and data engines in addition to measuring video quality (Shang et al. 2026). Despite this progress, most methods condition video generation through direct action projection, cross-attention, or action-conditioned modulation, leaving the resulting robot-environment consequences to be learned implicitly from the video-generation objective. LJM retains direct action conditioning while introducing a structured latent representation of how commanded motions change the robot and its environment.

Latent Reasoning for Embodied Dynamics

Vision-language-action models extend pretrained vision-language models to robot control by grounding language instructions and visual observations into action generation (Kim et al. 2024). To move beyond direct observation-to-action mapping, methods introduce reasoning-before-acting paradigms. ECoT performs embodied chain-of-thought reasoning over plans, sub-tasks, motions, and visual grounding cues, while CoT-VLA predicts future visual observations as intermediate reasoning steps for manipulation (Zawalski et al. 2024; Zhao et al. 2025). Although explicit reasoning can improve interpretability and planning, it may introduce inference overhead and restrict physical reasoning to linguistic or reconstructed visual spaces. Latent reasoning methods instead move intermediate reasoning into continuous representations. LaST₀ models future visual dynamics, 3D structure, and proprioceptive states through a latent spatio-temporal chain-of-thought, while LaRA-VLA internalizes multimodal chain-of-thought reasoning into latent space (Liu et al. 2026; Bai et al. 2026). These works show that continuous latent representations can capture fine-grained robotic dynamics that are difficult to express through language. However, they primarily use latent reasoning to improve policy-side action generation. In contrast, LJM predicts a sequence of temporally structured robot-environment interaction latent tokens from observations, instructions, and future actions, and shares these latents with a video diffusion expert to guide action-consistent future generation, extending latent reasoning from *reason-before-act* to *reason-before-render*.

Method

Problem Formulation

We formulate action-conditioned world modeling for robotic manipulation as predicting future visual observations given the current scene and future robot behaviors. Given a natural language instruction ℓ , a current RGB observation o_t , and a future robot action sequence $\mathbf{a}_{t:t+H-1}$ over horizon H , the goal is to generate the corresponding future observation sequence $\mathbf{o}_{t+1:t+H}$:

$$p_{\theta}(\mathbf{o}_{t+1:t+H} | o_t, \ell, \mathbf{a}_{t:t+H-1}).$$

Direct conditioning is insufficient because actions specify control commands but not their physical consequences; we therefore introduce an intermediate interaction representation. To this end, we propose a *reason-before-render* paradigm, which introduces an interaction-aware latent representation between action conditions and visual generation:

$$p_{\theta}(\mathbf{o}_{t+1:t+H} | o_t, \ell, \mathbf{a}_{t:t+H-1}, \mathbf{Z}_{\phi}),$$

where

$$\mathbf{Z}_{\phi} = f_{\phi}(o_t, \ell, \mathbf{a}_{t:t+H-1})$$

denotes the latent interaction representation produced by a dedicated latent reasoning expert. Unlike raw action embeddings, $\mathbf{Z}_{\phi}$ jointly captures robot motion, action-induced visual dynamics, and scene evolution, providing a structured prior to guide physically consistent video generation.

Model Architecture

As shown in Figure 2, we propose a Latent-Joint-conditional Model adopting a Mixture-of-Transformers (MoT) (Liang et al. 2025) architecture. To explicitly decouple robot-environment interaction modeling from visual synthesis, we construct our model with two distinct components: a latent reasoning expert based on Qwen3-VL-2B (Bai et al. 2025) and a world modeling expert based on the Wan2.2-5B (Wan et al. 2025) Diffusion Transformer. For the latent reasoning expert, given a language instruction ℓ , a current visual observation o_t , and a future robot action sequence $\mathbf{a}_{t:t+H-1}$ embedded by a lightweight MLP projector, it extracts continuous interaction latent tokens $\mathbf{Z}_{\phi} \in \mathbb{R}^{N_z \times d_q}$, where N_z is the number of latent tokens, and d_q is the latent reasoning expert hidden dimension. The N_z latent slots are appended after the context and action tokens following the temporal order of the future trajectory, and are decoded with the causal mask of the Qwen backbone. Each slot thus depends on the full conditioning and all preceding slots, $z_i = f_{\phi}(\mathbf{c}_t, z_{<i})$, computed in a single forward pass without teacher forcing. Importantly, the interaction tokens used for latent supervision are extracted before MoT Shared Attention. These tokens encapsulate the robot-environment interaction latent space—an interleaved representation of future visual dynamics, future action-induced flow, and future robot proprioception. This latent interaction knowledge is explicitly injected into the generation process of the world modeling expert through MoT Shared Attention, ensuring that the synthesized future frames are both visually coherent and strictly bound by robot-environment interactions. For the world modeling expert, it functions as a physical renderer to predict the corresponding

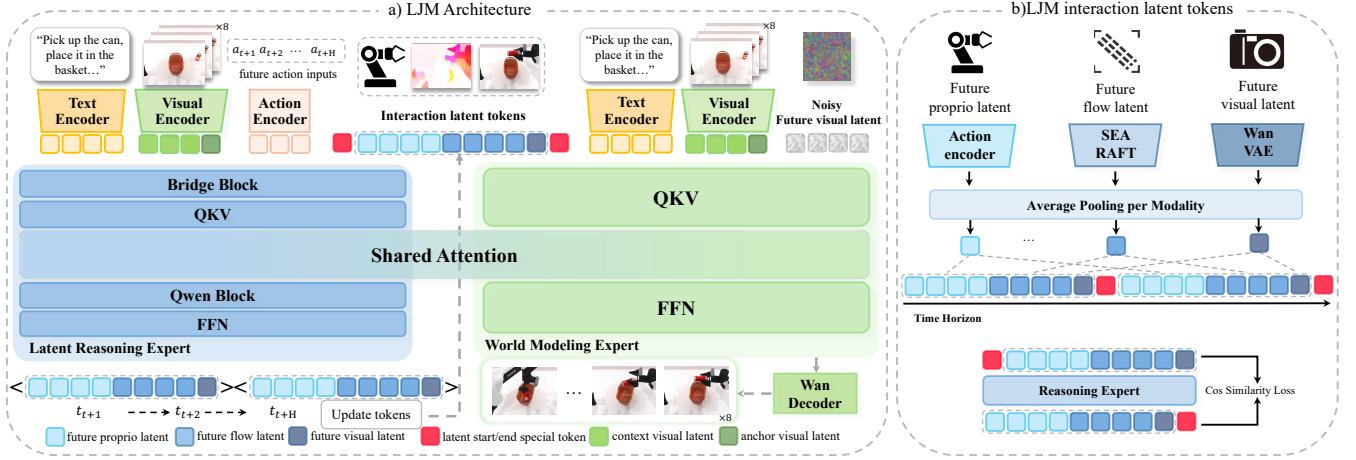


Figure 2: **Overview of the LJM architecture.** (a) The latent reasoning expert encodes language, visual context, anchor observation, and future actions into action-aware CoT latents, which are bridged into the video diffusion expert through MoT Shared Attention to guide future video generation. (b) During training, CoT latents are supervised by a unified target sequence constructed from future proprioceptive states, future flow latents, and future visual latents, encouraging them to encode robot–environment interaction dynamics.

future dynamics evolution. It takes the noisy target video latents $\mathbf{x}_\tau$ at diffusion timestep τ and predicts the denoising targets under the supervision of the ground-truth future video latents. To achieve better action control, we inject the robot actions at two specific locations within the world modeling expert: the cross-attention module and the AdaLN layer. The former provides action-conditioned contextual information, and the latter modulates the denoising dynamics.

Learning Action-conditioned Interaction Latents

The key challenge of action-conditioned world modeling is to bridge low-dimensional robot commands and their high-dimensional visual consequences. To learn such an intermediate interaction space, we supervise the latent reasoning expert with future observations that explicitly describe three complementary aspects of robot–environment evolution: scene appearance changes, action-induced motion, and robot physical trajectories.

Unified interaction target construction. Given future observations, we construct a unified latent target sequence from three complementary aspects of robot–environment evolution: visual dynamics, motion dynamics, and proprioception.

For visual dynamics, the ground-truth future video is encoded by the Wan VAE into spatio-temporal latent features $\mathbf{Z}^v$. We spatially aggregate and temporally compress these features to obtain compact visual dynamics tokens:

$$\mathbf{Y}^v = p_{\text{temp}} \left(\frac{1}{HW} \sum_{h,w} \mathbf{Z}_{:,h,w}^v \right),$$

where $p_{\text{temp}}(\cdot)$ denotes temporal pooling. These tokens capture future scene evolution, including object displacement and state transitions.

For robot dynamics, we encode future end-effector states $\mathbf{s}_{t+1:t+H}$ through the same embedding module as the input

action sequence:

$$\mathbf{Y}^s = m_s(\mathbf{s}_{t+1:t+H}),$$

providing explicit supervision on the future physical trajectory of the robot.

To capture visual motion dynamics, we extract motion cues from future observations using dense optical flow. Specifically, optical flows between consecutive frames are precomputed by SEA-RAFT (Wang, Lipson, and Deng 2024). For each temporal transition, the corresponding flow fields are accumulated over the sampling interval:

$$\mathbf{F}_{t+k} = \sum_i \text{SEA-RAFT}(o_i, o_{i+1}),$$

where the summation reduces to adjacent-frame optical flow when the sampling interval is one. The resulting flow fields are summarized into compact motion descriptors $\mathbf{G}_{t+k}$ and projected into the latent space:

$$\mathbf{Y}^f = m_f(\mathbf{G}_{t+1:t+H}),$$

where m_f consists of a LayerNorm followed by a linear projection. These motion tokens provide explicit supervision on temporal displacement and visual dynamics, complementing appearance-level visual targets and robot-state targets.

Temporal chunk-wise target organization. The three modalities describe different aspects of the same future evolution process, but their temporal resolutions are different. Specifically, proprioceptive and flow targets are defined at the robot control frequency, while visual targets are obtained after temporal compression by the Wan VAE. To preserve the temporal relationship among these modalities, we organize target tokens in a chunk-wise manner instead of independently concatenating each modality.

For the j -th visual token, we first associate it with the corresponding temporal interval:

$$s_j = \left\lfloor \frac{jH}{T_v} \right\rfloor, \quad e_j = \left\lfloor \frac{(j+1)H}{T_v} \right\rfloor.$$

Within each temporal chunk, proprioceptive and flow tokens are arranged chronologically, followed by the corresponding pooled visual token:

$$\mathbf{Y}^{(j)} = [\mathbf{Y}_{s_j}^s, \mathbf{Y}_{s_j}^f, \dots, \mathbf{Y}_{e_j-1}^s, \mathbf{Y}_{e_j-1}^f, \mathbf{Y}_j^v].$$

The final unified target sequence is obtained by concatenating all temporal chunks:

$$\mathbf{Y} = \text{ChunkInterleave}(\mathbf{Y}_{1:H}^s, \mathbf{Y}_{1:H}^f, \mathbf{Y}_{1:T_v}^v).$$

For example, with $H = 16$ and $T_v = 4$, each visual token corresponds to four robot timesteps, resulting in $16 + 16 + 4 = 36$ latent target tokens. This organization preserves the temporal causality between robot motion, induced visual dynamics, and resulting scene evolution.

Training

The video diffusion expert is trained with the standard diffusion flow-matching loss $\mathcal{L}_{\text{video}}$. Independently, let Ω denote the valid visual, flow, and proprioceptive target positions. We align the predicted interaction latent tokens with the detached target sequence using masked cosine distance:

$$\mathcal{L}_{\text{cos}} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \left(1 - \frac{\mathbf{Z}_{\phi,i}^\top \mathbf{Y}_i}{\|\mathbf{Z}_{\phi,i}\|_2 \|\mathbf{Y}_i\|_2} \right).$$

We additionally regularize the valid predicted tokens with a variance term that prevents dimensional collapse and a covariance term that discourages redundant dimensions. The latent reasoning expert is optimized with $\mathcal{L}$:

$$\begin{aligned} \mathcal{L}_{\text{latent}} &= \mathcal{L}_{\text{cos}} + 0.02 \mathcal{L}_{\text{var}} + 0.01 \mathcal{L}_{\text{cov}}, \\ \mathcal{L} &= \mathcal{L}_{\text{video}} + 0.4 \mathcal{L}_{\text{latent}}. \end{aligned}$$

We use a two-stage training schedule: First, we pretrain LJM on the DROID dataset (Khazatsky et al. 2024). Following Ctrl-World (Guo et al. 2025), we spatially concatenate synchronized camera views and retain both successful and failed trajectories during pretraining. To expose LJM to physical interaction dynamics in both simulation and the real world, we further fine-tune LJM on expert demonstrations, policy rollouts, and failure trajectories collected from LIBERO (Liu et al. 2023a), RoboTwin (Chen et al. 2025), and our in-house Franka platform, which consists of a Franka R3 robot arm equipped with a Robotiq gripper. To support cross-embodiment training, we use a unified 14-D action space with end-effector pose displacements in SE(3) plus gripper state for each of two arm slots. We standardize each camera view to 320×240 resolution. Given 8 historical frames and 1 current frame, LJM conditions on the next 16 actions to generate 16 future video frames. We trained the model on 32 PPU-ZW810E with a global batch size of 32. More details of the dataset and training can be found in the Appendix.

Experiments

In this section, we conduct experiments to evaluate LJM in terms of visual quality, policy evaluation and policy improvement capability. We aim to answer the following questions: (1) Can LJM generate action-conditioned rollouts with high visual quality and temporal consistency? (2) Can LJM reliably evaluate generalist robot policies through closed-loop rollouts by faithfully modeling robot motion, object interactions, and scene evolution? (3) Can LJM improve policy performance by synthesizing successful corrective trajectories from real-world failures?

World Model Visual Quality Analysis

Benchmarks and Baselines We evaluate visual quality on two benchmarks. LIBERO (Liu et al. 2023a) is a widely used robotic manipulation benchmark featuring diverse tasks with rich robot–object interactions and scene dynamics. For evaluation, we hold out 10% of the trajectories from the four LIBERO suites as the validation set and randomly sample 100 episodes for testing. We compare the predicted videos with their ground-truth counterparts using both computational metrics, including PSNR (Hore and Ziou 2010) and SSIM (Wang et al. 2004), and model-based metrics, including LPIPS (Zhang et al. 2018) and FVD (Unterthiner et al. 2019). WorldArena (Shang et al. 2026) evaluates embodied world modeling using EWMScore and a suite of fine-grained metrics covering visual fidelity, scene consistency, geometry, motion dynamics, semantic alignment, instruction following, and interaction quality.

LIBERO As shown in Table 1, we compare LJM with IRASim (Zhu et al. 2025a), Ctrl-World (Guo et al. 2025), Wan2.2-TI2V (Wan et al. 2025), and WoVR (Jiang et al. 2026), covering different pretrained backbones and action-conditioning mechanisms. Notably, WoVR uses the same backbone and AdaLN–Cross-Attention conditioning as LJM but without MoT Shared Attention. LJM achieves the best results on all four metrics, improving PSNR and SSIM by 1.35 and 0.0418 while reducing FVD and LPIPS by 30.59 and 0.0391 over the strongest baseline. The latent CoT ablations further show that removing flow, video, or proprioceptive supervision consistently degrades generation quality, demonstrating their complementary roles. Qualitative comparisons are shown in Figure 3(a).

WorldArena As shown in Table 2, we compare LJM with representative embodied world models and general video generation models, including IRASim (Zhu et al. 2025a), WoVR (Jiang et al. 2026), Ctrl-World (Guo et al. 2025), Wan2.2 (Wan et al. 2025), WoW (Chi et al. 2025), and Genie Envisioner-Sim 2.0 (Qiu et al. 2026), covering diverse model architectures and action-conditioning designs. LJM achieves the highest EWMScore among the compared methods, reaching 73.06 and outperforming Genie Envisioner by 3.47 points. LJM performs particularly well on Dynamic Degree (Teed and Deng 2020), Flow Score (Liu et al. 2023b), and Motion Smoothness (Zhang et al. 2024), and ranks second on Image Quality (Ke et al. 2021), Instruction Following (Bai et al. 2025), and JEPA Similarity (Bardes et al.

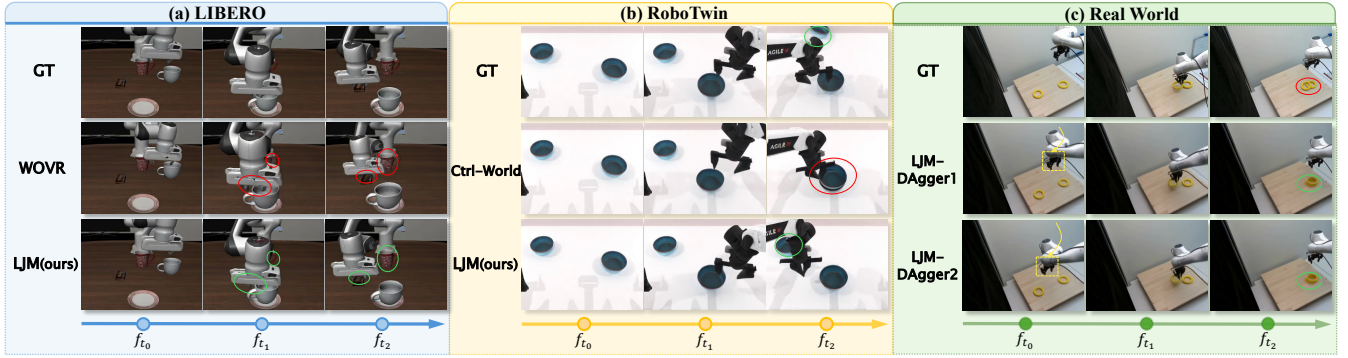


Figure 3: **Qualitative rollouts across simulation and real-world settings.** (a) On LIBERO, LJM better preserves robot appearance and action-conditioned interactions than WoVR. (b) In a failed RoboTwin policy rollout, LJM reproduces the failure under closed-loop policy interaction, whereas Ctrl-World incorrectly predicts a successful outcome. (c) On our real-world Franka platform, LJM-Dagger generate successful corrective trajectories from perturbed initial states. Yellow arrows indicate end-effector perturbations, while green circles mark successful outcomes.

Method	PSNR $\uparrow$	SSIM $\uparrow$	FVD $\downarrow$	LPIPS $\downarrow$
IRASim	24.92	0.8713	287.50	0.0639
Ctrl-World	24.78	<u>0.8820</u>	<u>77.09</u>	0.0670
Wan2.2-TI2V	15.33	0.6451	653.82	0.2640
WoVR	<u>27.25</u>	0.8666	180.97	0.0763
LJM (Ours)	28.60	0.9238	46.50	0.0248
<i>w/o Flow</i>	27.50	0.9125	57.96	0.0310
<i>w/o Video</i>	27.86	0.9170	47.24	0.0275
<i>w/o Proprioception</i>	27.57	0.9140	52.23	0.0289

Table 1: LIBERO visual quality and latent tokens ablations on 100 test episodes. Each ablation removes one target from full LJM. Best and second-best results are bolded and underlined.

Method	EWMScore	Motion	Dynamic	Flow
IRASim	59.63	64.31	21.60	11.66
WoVR	61.68	69.74	22.43	12.55
Ctrl-World	63.72	77.34	41.82	33.57
Wan2.2	62.35	76.45	31.20	18.06
WoW	65.53	<u>87.29</u>	<u>53.04</u>	<u>60.23</u>
GE-Sim 2.0	<u>69.59</u>	83.08	51.34	52.07
LJM (Ours)	73.06	91.42	76.27	99.82

Table 2: WorldArena results. Due to space constraints, only a subset of metrics is shown; full results are in the Appendix.

2023). These results highlight LJM’s ability to reason about action-driven motion and temporally coherent robot–object interactions in the latent space.

World Model as Policy Evaluator

Visual quality alone is insufficient to reflect whether a world model faithfully captures the interactions between policy actions and the world. We therefore evaluate whether LJM can serve as a reliable policy evaluator through policy-in-the-loop rollouts while preserving relative policy performance across repeated robot executions.

Tasks and Policies. We consider two strong policy baselines representing current two prominent paradigms in generalist robot learning: the Vision-Language-Action model $\pi_{0.5}$ (Physical Intelligence, Black et al. 2025) and the World Action Model LingBot-VA (Li et al. 2026). We evaluate both policies on 10 manipulation tasks. The five real-world Franka tasks are *Stack Ring*, *Stack Jenga*, *Hang on Cup*, *Hang on M*, and *Put Chili in Drawer*. The five RoboTwin bimanual tasks are *Stack Bowls*, *Turn Switch*, *Place Can in Basket*, *Move Stapler to Pad*, and *Move Can to Pot*.

Metrics. The primary metric is the success rate for each task–policy pair. Following SIMPLER (Li et al. 2024) and Ctrl-World (Guo et al. 2025), we assess agreement between world model and real rollouts using Pearson correlation r , the least-squares regression equation, and Mean Maximum Rank Violation (MMRV). We compute these statistics separately for the RoboTwin bimanual tasks, the real-world Franka tasks, and all tasks jointly. We additionally report task-wise policy-ranking accuracy, defined as the fraction of tasks for which world model evaluation identifies the same better-performing policy as real evaluation.

Qualitative and Quantitative Results. Figure 3(b) presents qualitative comparisons between LJM and Ctrl-World, while Figure 4(a) visualizes the all task–policy evaluation results. On RoboTwin, LJM achieves $r = 0.872$, $y = 0.98x - 0.08$, a mean task-wise MMRV of 0.000, and 100% ranking accuracy. On real-world Franka tasks, LJM obtains $r = 0.748$, $y = 0.66x + 0.29$, an MMRV of 0.015, and 80% ranking accuracy. Overall, LJM reaches $r = 0.785$, an MMRV of 0.008, and 90% ranking accuracy, substantially outperforming Ctrl-World at 0.450, 0.118, and 40%, respectively. To ensure accurate evaluation, multiple annotators judge success from anonymized rollout videos. The complete evaluation criteria are provided in the Appendix.

Analysis of these results reveals that, despite visually convincing generations, Ctrl-World often predicts success for real failures, producing false positives. This indicates that visual fidelity alone does not guarantee faithful modeling of

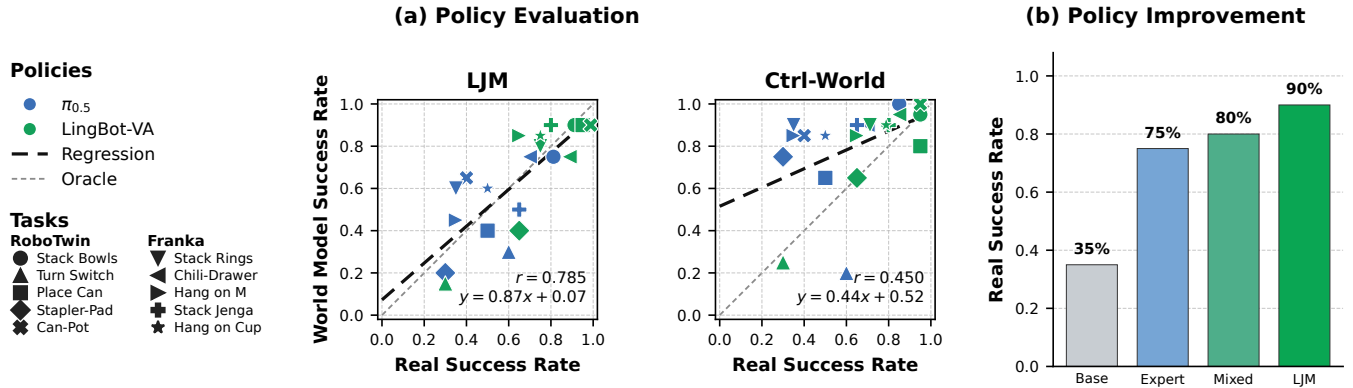


Figure 4: **(a) Policy evaluation.** Correlation between real and world-model success rates across five RoboTwin and five real-world Franka tasks. Both LJM and Ctrl-World are evaluated with $\pi_{0.5}$ and LingBot-VA. Dashed lines are least-squares fits and the gray diagonal is the oracle $y = x$. **(b) Policy improvement.** Real-world success rates of the base $\pi_{0.5}$ and policies post-trained on 100 additional expert demonstrations, a 50/50 mixture of expert and LJM-Dagger trajectories, and 100 LJM-Dagger trajectories.

policy actions and robot-object interactions, which LJM addresses through explicit latent reasoning. Additional results and visualizations are provided in the Appendix.

World Model for Policy Improvement

Beyond closed-loop policy evaluation, we investigate whether LJM-generated synthetic data can improve a real policy. Inspired by DAGger’s principle of aggregating corrective supervision at states encountered by the learned policy (Ross, Gordon, and Bagnell 2011), we introduce *LJM-Dagger*: (1) We collect failed rollouts from the base $\pi_{0.5}$ policy across five real-world tasks and perturb their initial states with short action in LJM, producing diverse initial observations and corresponding end-effector poses. (2) We use multi-view geometry to localize the manipulated object and task goal in the world frame, retrieve successful demonstrations with nearby goal positions, and retarget their trajectories to each perturbed state. LJM renders the videos for resulting candidate trajectories, from which expert annotators retain successful rollouts, as visualized in Figure 3(c). More details of algorithm is provided in the Appendix.

Real-world Results. We evaluate on *Stack Ring*, where the base $\pi_{0.5}$ achieves its lowest success rate of 35%. The task requires precisely stacking one ring-shaped object on another without toppling, making it a stringent test of manipulation accuracy and LJM-Dagger. As shown in Figure 4(b), expert-only, mixed, and LJM-Dagger-only post-training achieve 75%, 80%, and 90% success, respectively. Surprisingly, fully synthetic training improves the base policy by 55% points and exceeds expert-only training by 15%. Our analysis suggests that failure-conditioned perturbations broaden the manipulation state space beyond expert demonstrations collected from similar initial states. The results demonstrate that LJM’s action following and object-interaction modeling can transform policy failures into effective corrective supervision.

Conclusion

We presented LJM, a Latent Joint-conditional Model for action-conditioned world modeling. LJM introduces a reason-before-render paradigm that uses MoT Shared Attention to integrate interaction latent tokens capturing future visual, optical-flow, and proprioceptive dynamics. Across LIBERO and WorldArena, LJM achieves the best visual quality among the compared action-conditioned world models, spanning diverse video backbones and action-conditioning mechanisms. We further evaluate LJM through closed-loop policy evaluation and improvement. LJM rollouts achieve a Pearson correlation of 0.785 with real policy performance and correctly rank policies on nine of ten tasks. Moreover, LJM-Dagger transforms failed executions into corrective synthetic trajectories, improving real-world policy success rate from 35% to 90%. These results demonstrate that LJM supports not only high-quality embodied video generation, but also practical policy evaluation and improvement.

Limitations and Future Work

Despite the promising results, several limitations remain. One limitation is multi-view consistency. As described in Training section, following Ctrl-World (Guo et al. 2025), LJM concatenates synchronized camera views without explicitly enforcing cross-view geometric or motion consistency. Future work will explore camera-aware generation like 4D RoPE, view-label conditioning, and cross-view consistency objectives. Another limitation is that precise contact, deformable objects, and long-horizon interactions remain constrained by the video backbone; richer latent reasoning with 3D geometry and contact information may improve physical modeling. Finally, the complexity of generated rollouts makes data quality in policy evaluation and improvement difficult to assess using rule- or model-based methods. We plan to explore more reliable automatic evaluation and filtering in the future.

References

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